

BU 201: Introduction to Theory of Structures

SHEAR FORCE AND BENDING MOMENTS

Introduction: When any structure is loaded, stresses are induced in the various parts of the structure, and in order to calculate the stresses, where the structure is supported at a number of points, the bending moments and shearing forces acting must also be calculated. In general, a structure may be considered to consist of a series of beams linked together in some way and further, the complete structure may be treated as a beam with an elaborate cross-section. Calculations can be made progressively first on the structure as a whole and then on the individual parts.

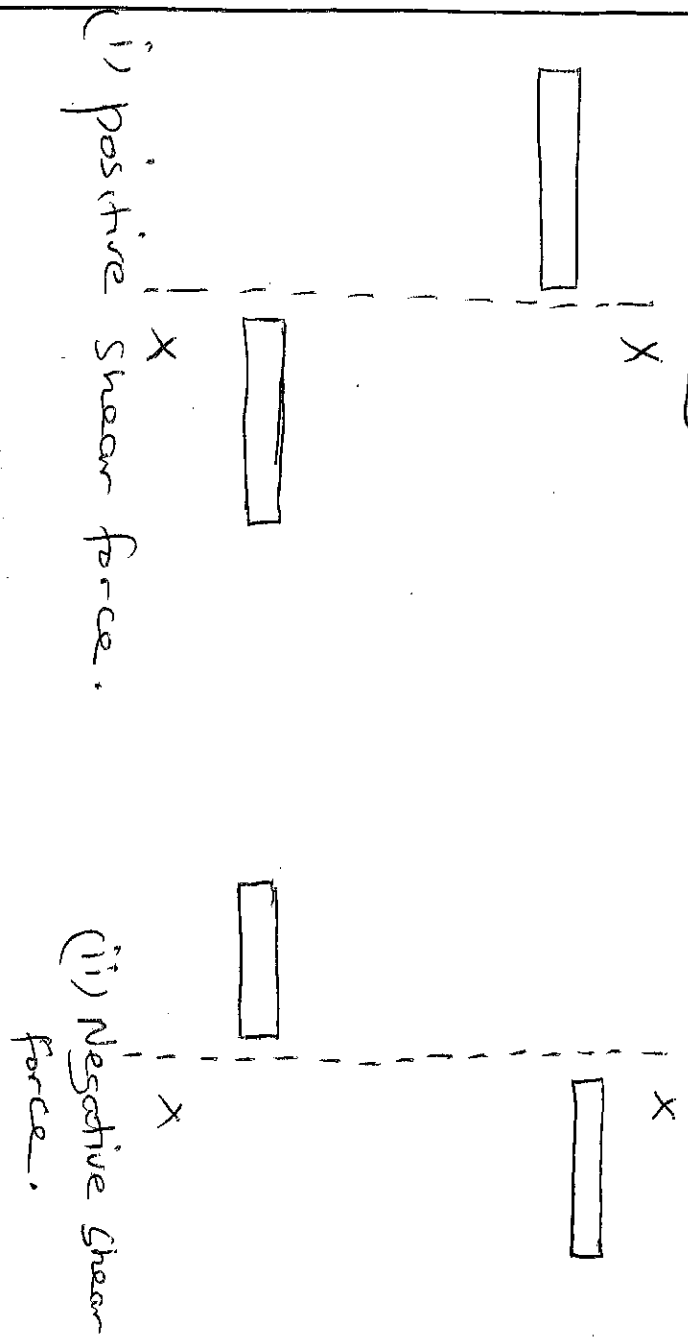
SHEARING FORCE: The shearing force ($S.F$) at the section is defined therefore as the algebraic sum of the forces taken on one side of the section.

BENDING MOMENT ($B.M$): Is defined as the algebraic sum of moments of the force about the section taken either side of the section.

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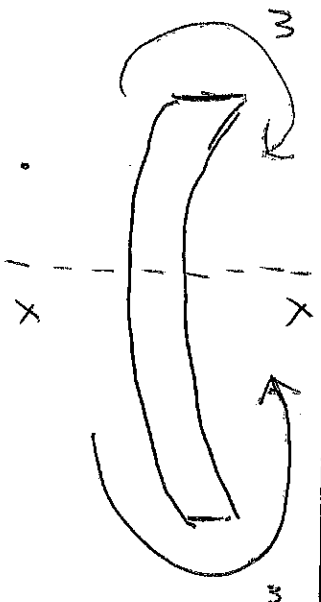
SIGN CONVENTION FOR SHEARING FORCE:

A shearing force having an upwards direction to the right hand side of a section or downwards to the left of the section will be taken positive, Similarly, a negative shearing force will be one that has a downward direction to the right of the section or upward direction to the left of the section. Thus, it can be illustrated diagrammatically as shown below:

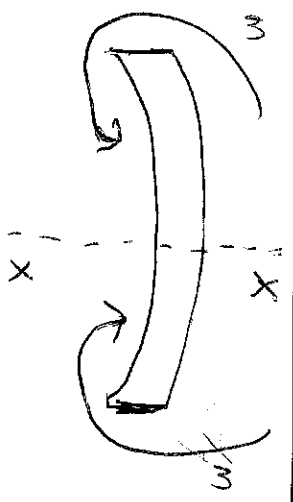


SIGN CONVENTION FOR BENDING MOMENT

A bending moment causing concavity upwards will be taken as positive and called as sagging bending moment, a bending moment causing upwards will be taken as negative and called hogging bending moment.



(i) Positive bending moment.



(ii) Negative bending moment.

LOADS

load is defined as anything that has the tendency of generating internal stress in a structure, which tends to deform the structure.

Types of load

(i) Dead load: Dead loads are loads due solely to the weight of the structure itself and can usually be estimated quite accurately.

(ii) Live loads: These are structures referred to as imposed loads, live load of a building is the sum of all other loads on the structure which may arise from a wide variety of sources such as: occupants and furniture. This type of load can only be estimated as they vary from time to time during the normal working condition of the building.

CLASSIFICATION OF LOADS

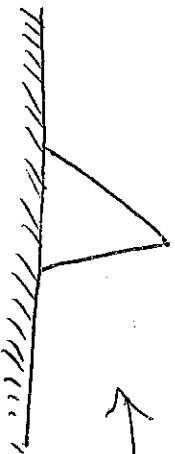
loads can also be classified as:

- (i) Static and Dynamic load: Static loads are loads that are expected to remain constant during their life span, where as dynamic loads are loads that varies significantly over a relative short time, so that the response of the structure to the loading is affected by resonance.
- (ii) Direct loads and indirect loads: Direct loads are external loads applied to the structure such as loads of occupants and furniture, where as indirect loads are loads on a structure due to loads not directly applied. Such as expansion and contraction due to variation in temperature.

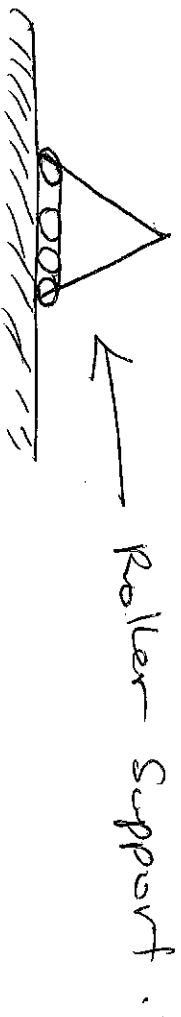
- (iii) Point (concentrated) load and Distributed load: Point load is load assumed to act at a point where as distributed load is distributed over the surface of the member per unit length per area. If the value of the distributed load is constant over the length or one in which it acts is termed as uniformly distributed load (UDL).

Types of Support

(i) Pinned Support: A pinned support is one which prevents the structure from moving translationally in any direction at the support point, but ~~prevents~~ provides no resistance to rotation. It provides no resistance to rotation. It provides degree of restraints. The symbol used to represent pinned support is shown below.

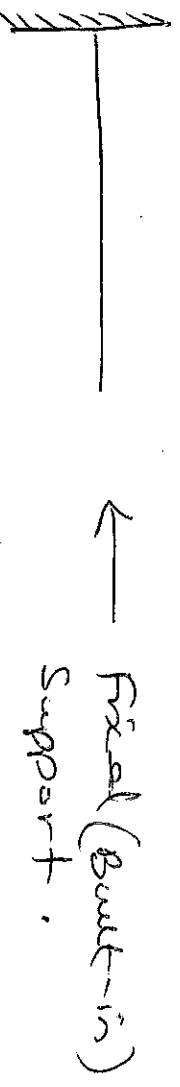


(ii) Roller Support: A roller support allows some translation as well as rotational movements. Roller support prevents vertical translation of the structure at the support point but allows both horizontal translation and rotation at the support point. It provides only one degree of restraint. The symbol used to represent roller support is shown below.



(iii)

Fixed (Built-in) Support: This prevents all movements of the structure at the support point both translations and rotations. It provides three-degree of restraints. The symbol used to represent fixed support is shown below.



EQUATIONS OF EQUILIBRIUM

A system of forces is said to be in static equilibrium if the sums of all the forces and of all the moments are equal to zero at any point.

In statics, all systems must satisfy the requirements of equilibrium, if equilibrium is not maintained, then out of balance, force will cause acceleration, so that the system is no longer behaving statically.

For a three-dimensionally system with a set of mutually perpendicular axes x, y , and z as shown below, six (6) conditions must be satisfied. The force components in each direction must sum to zero, and the moments of the forces about each axis must sum to zero. These can be expressed

Mathematically as:

$$\sum F_x = 0$$

Where $\sum F_x$ is the Component of any

$$\sum F_y = 0$$

force in the x-direction.

$$\sum F_z = 0$$

$\Rightarrow F_y$ is the Component of any force in the y-direction,

$$\sum M_x = 0$$

$\Rightarrow F_z$ is the moment of a force about the z-direction,

$$\sum M_y = 0$$

$\Rightarrow M_x$ is the moment of a force about the x-axis

$\Rightarrow M_y$ is the moment of a force

about the y-axis

$\Rightarrow M_z$ is the moment of a force

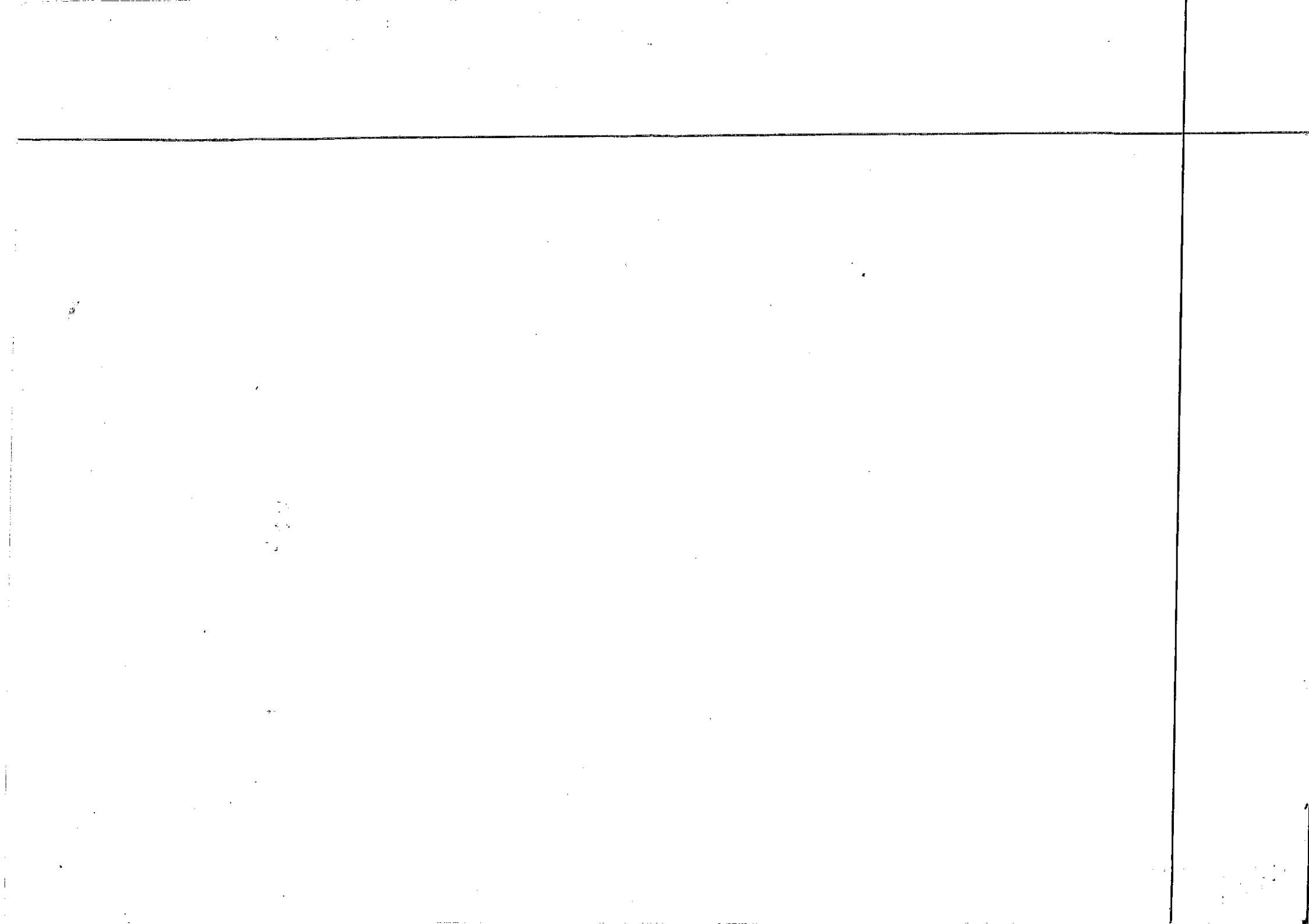
about the z-axis,

For a two-dimensional system, where the system is taken to lie entirely on one plane (x-y), it can be expressed in terms of force Component in the x and y directions and moment about the z-axis. The condition for equilibrium is as shown belows:

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum M_z = 0$$



1.5 RELATIONSHIP BETWEEN LOAD, SHEAR FORCE AND BENDING MOMENT.

The beam shown in figure 1(a) below carries distributed loading which varies in an arbitrary manner $f(w)$. Consider the free body (figure 1(b)); of a small slice of length dx for which the loading may be regarded as uniform, w .

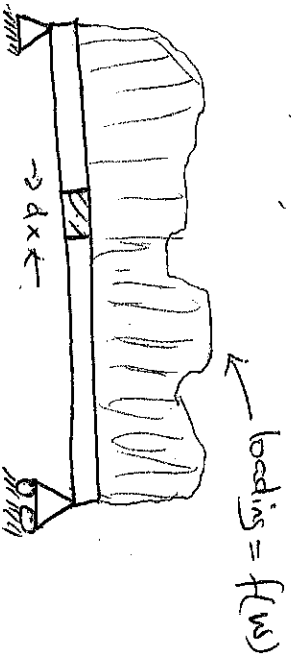


Figure 1(a)

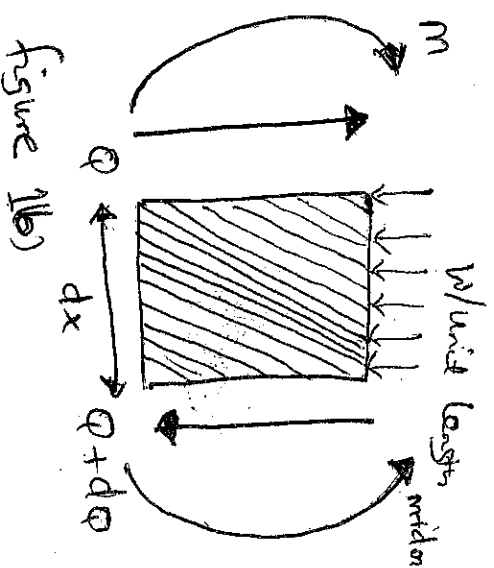


figure 1(b)

for vertical equilibrium

$$-Q + wdx + (Q + dQ) = 0$$

$$\Rightarrow wdx + dQ = 0$$

$$w dx = -dQ$$

$$w = -\frac{dQ}{dx} \quad \text{--- Eq. (1)}$$

For: Moment Equilibrium

Taking moments about the right-hand edge.

$$-M - Qdx + wdx\left(\frac{dx}{2}\right) + (M + dM) = 0$$

$$-Qdx + \frac{w(dx)^2}{2} + dM = 0$$

$$\text{where } (dx)^2 = 0$$

$$\Rightarrow -Qdx + \frac{w(dx)^2}{2} + dM = 0$$

$$\therefore -Qdx + dm = 0$$

$$\Rightarrow -Qdx = -dm$$

$$+Q = \frac{dm}{dx}$$

$$Q = \frac{dm}{dx} \quad \text{--- Eq. (2)}$$

Substituting equation (2) in equation (1), it becomes:

$$w = -\frac{d}{dx}\left(\frac{dm}{dx}\right)$$

$$w = \frac{d^2 m}{dx^2} \quad \text{--- Eq. (3)}$$

From equation (1) we have,

$$dQ = -w dx$$

when we integrate

$$\int_1^2 dQ = \int_1^2 -w dx$$

$$[Q]_1^2 = \int_1^2 -w dx + A \quad \text{--- (4)}$$

$$Q_2 - Q_1 = \int_1^2 -w dx + A$$

Thus the change in shear force between any two cross-sections may be obtained from the area under the load distribution curve between those sections.

From equation (2) we have,

$$dm = Q dx$$

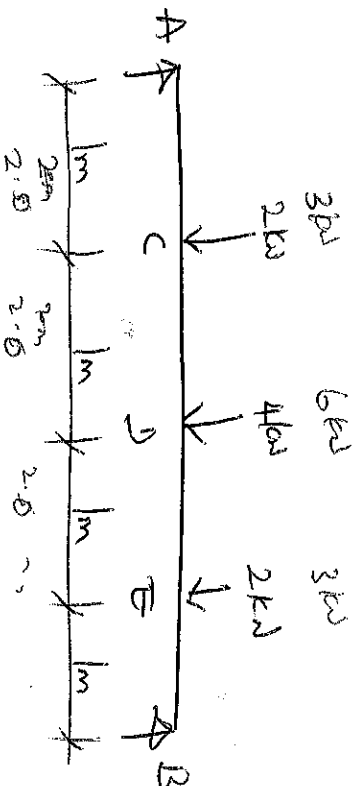
$$\int_1^2 dm = \int_1^2 Q dx$$

$$[m]_1^2 = \int_1^2 Q dx + B$$

$$M_2 - M_1 = \int Q dx + B \quad \text{--- eq. (5)}$$

Thus the change in bending moment between any two sections is found from the area under the shear force diagram between those sections.

Example: Draw the S.F and B.M diagrams for Simply Supported beam loaded as shown below



Soln: To determine the Support reactions taking Moments about A, we get

$$R_B \times 10 = 2 \times 1 + 4 \cdot (1+1) + 2(1+1+1+1)$$

$$= 2 + 8 + 6$$

$$= 16$$

$$R_B = \frac{16}{4}$$

$$R_B = \underline{\underline{4kN}}$$

But $R_A + R_B = 2 + 4 + 2$

$$R_A + R_B = 8kN$$

but $R_B = 4kN$

$$\therefore R_A + 4kN = 8$$

$$R_A = 8 - 4$$

$$R_A = \underline{\underline{4kN}}$$

Shear Force Calculations:

$$S_{B-E} = +4 \text{ kN}$$

$$S_{E-D} = 4 - 2 = 2 \text{ kN}$$

$$S_{D-C} = 2 - 4 = -2 \text{ kN}$$

$$S_{C-A} = -2 - 2 = -4 \text{ kN}$$

Bending Moment Calculations:

$$M_B = 0$$

$$M_E = 4 \times 1 = 4 \text{ kNm}$$

$$M_D = 4(1+1) - 2 \times 1 = 8 - 2 = 6 \text{ kNm}$$

$$\begin{aligned} M_C &= 4(1+1+1) - 2(1+1) - (4 \times 1) \\ &= 12 - 4 - 4 = 4 \text{ kNm} \end{aligned}$$

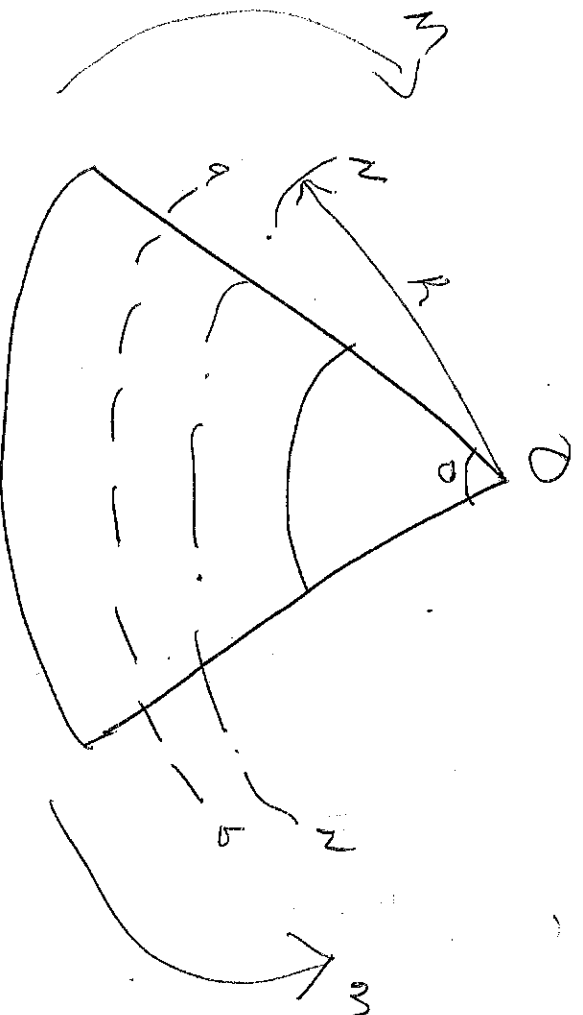
$$\begin{aligned} M_A &= 4(1+1+1+1) - 2(1+1+1) - 4(1+1) - 2 \times 1 \\ &= 16 - 6 - 8 - 2 = 0 \text{ kNm} \end{aligned}$$

Principles of Simple Bending.

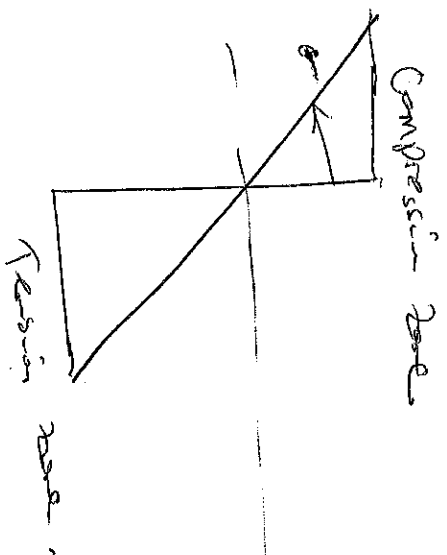
Pure Bending: If a length of a beam is acted upon by a constant moment (zero shearing force), the stresses set up on any cross-section must constitute a pure couple equal in magnitude to the bending moment. Therefore, it can be deduced that one part of the cross-section is in compression and the other part is in tension.

① & ②

Consider the figure below: It is subjected to the condition that the end sections remain plane, it is clear that for an initially straight beam the inside or concave edge will be in compression and the outside or convex edge will be in tension. There will be an intermediate surface at which the stress is zero (neutral surface), the neutral surface cuts any cross-section in the neutral axis.



①



Theory of Simple Bending (Bending Equation)

When a beam is loaded it is bent. A subjected to stresses bending moments. Consequently, longitudinal or bending stresses are induced in cross-section. In order to determine the practical utility of any beam, ~~the~~ bending it is necessary to establish a relationship between the radius of curvature to which the beam bends, the bending moment, the bending stress and its cross-sectional dimensions. The equation which connects these quantities is known as the bending equation.

ASSUMPTIONS IN THEORY OF BENDING

- (i) The material of the beam is perfectly homogeneous throughout.
- (ii) The stress induced is proportional to the strain and at no place the stress exceeds the elastic limit.

(iii) The value of modulus of elasticity (E) is same, for the fibres of the beam under compression or under tension.

(iv) The transverse section of the beam, which is plane before bending, remains plane after bending.

(v) There is no resultant pull or push on the cross-section of the beam.

(vi) The loads are applied in the plane of the bending.

(vii) The transverse section of the beam is symmetrical about a line passing through the centre of gravity in the plane of bending.

(viii) The radius of curvature of the beam before bending is very large in comparison to its transverse dimensions.

As a result of a bending moment or couple, a length of beam will take up a curved shape, and a very short length may be treated as a part of the arc of a circle. It follows that at the outer radii the material will be in tension and at the inner radii in compression, and at some radius there will be no stress.

This layer of the material is the neutral layer or neutral axis.

Figure 3: below shows

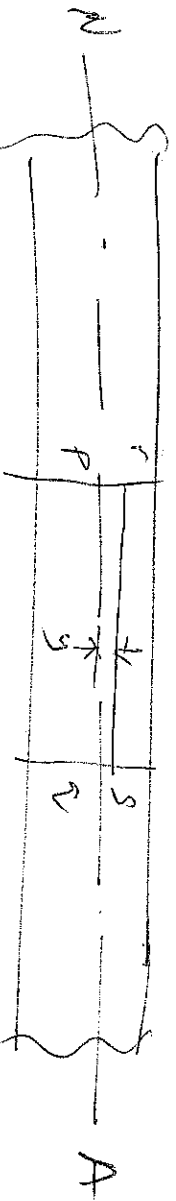
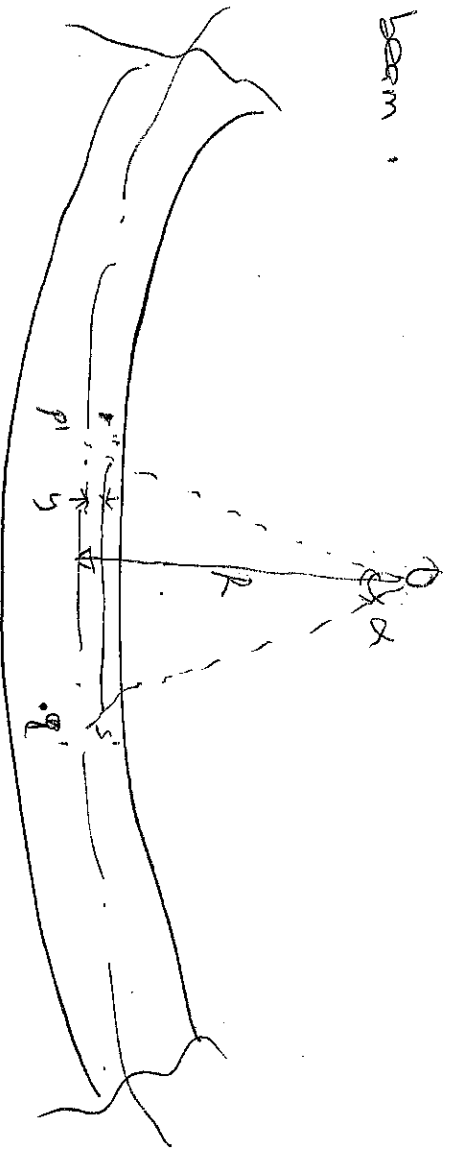


Figure 3: Shows a longitudinal section of a beam.



The neutral layer (axis) N.A. being bent to form an arc of a circle of radius R . The neutral layer is then, before bending, the length PQ , which after bending becomes $P'Q'$.

Consider some layer rs at a distance y from PQ , which after bending becomes $r's'$.

Let $P'Q'$ subtend an angle α at the centre of curvature.

$$P'Q' = R\alpha \quad \text{--- Eq. ①}$$

$$r's' = (R-y)\alpha \quad \text{--- Eq. ②}$$

④

Initially the parallel layers would have equal lengths,

$$p_g = r_s$$

At line there is no stress at the neutral axis or layer, there is no strain.

$$\rho'_{\alpha} = \rho_{\alpha}$$

$\frac{S}{H} = \frac{1}{\sqrt{1 + \frac{1}{(1 - \frac{1}{H})^2}}}$

but $r_s = p_q = p'_1 q'_1$

$\frac{5x^3 + x^2 - 1}{x-1} = 5x^2 + 6x + 7$

53
P
3
11
P
2

$$\frac{2}{5} \left(\frac{R-1}{2} \right)$$

$$\begin{array}{r} 5 \text{ st } 5: \\ 11 \\ R \\ R - (R - 5) \\ \hline R \\ R \end{array}$$

Now if the stress is $\sigma = b$

Then modulus E , then

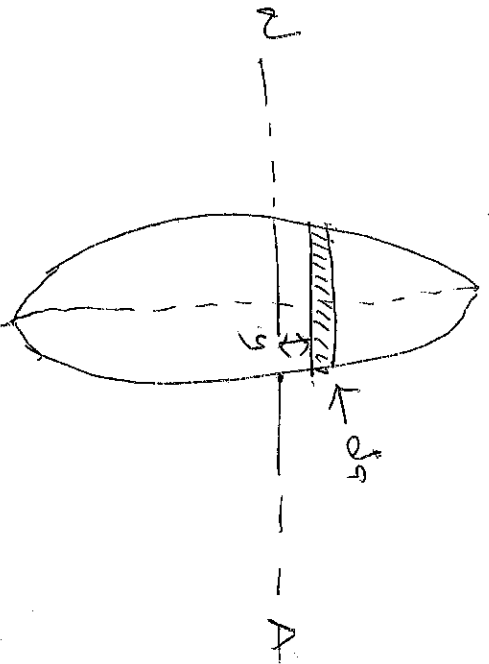
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17 | 10

7

if a transverse section of the beam is now
Consider figure 4 below:



Let a strip of area dy , lie at a distance y from the neutral axis.

Then, the normal force on this area

$$(dF) = \frac{E}{R} y dy \quad \text{--- eq (8)}$$

Now the moment of this force about the neutral axis is

$$= \frac{E}{R} y dy \times y \quad \text{or} \quad \text{--- eq (9)}$$

$\frac{E}{R} y^2 dy$ This is the resisting moment of

the material caused by the stress produced, & this the total resisting moment is

$$= \int \frac{E}{R} y^2 dy \quad \text{or} \quad \text{--- eq (10)}$$

(6)

And $E y^2 da$ is the second moment of area about the neutral axis, I_{NA} .

$\therefore$ Resisting moment $M = \frac{E}{R} \times I$
but since the resisting moment balances the applied bending moment.

$$\therefore M = \frac{E}{R} \times I \quad \text{or}$$

$$\frac{M}{I} = \frac{E}{R} \quad \text{but}$$

$$\frac{E}{R} = \frac{E y^2}{I}$$

$$\therefore \frac{M}{I} = \frac{E y^2}{I} = \frac{E}{R} \quad \text{--- Eq (11)}$$

~~Thus~~ $\therefore$ where $M =$ Moment of resistance

$I =$ moment of inertia of the section about neutral axis (m^4)

$E =$ Young's modulus of elasticity

$R =$ Radius of curvature of m^4

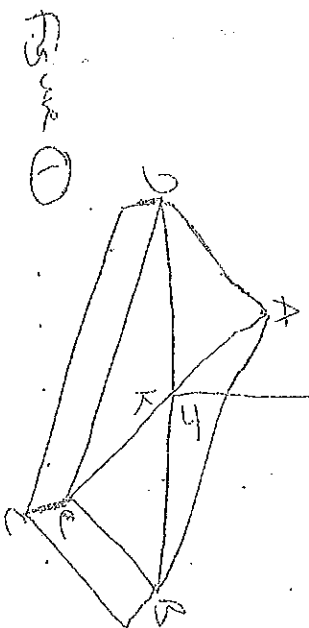
$\sigma =$ Bending Stress

The above equation is known as the Bending Equation. (11)

(7)

DETERMINATION OF CENTRE OF GRAVITY

The Centre of Gravity of some objects may be found by balancing the object on a point. Take a thin plate of thickness t , shown in figure (1) below. Draw the diagonals of the upper and lower faces to intersect at J and K respectively. If the plate is placed on point at K , the plate will not fall. That is, it is balanced. If suspended from J , the plate will hang horizontally. The Centre of Gravity of the plate is at the ~~middle~~ centre of the line JK .



Another method for determining the Centre of Gravity is by Suspension. Take an object, the Section of which is shown in figure (2) below. Suspend it from point L . The body will not come to rest until its resultant weight is vertically downward from L .

Through L , draw a vertical line LN . Then suspend the body from a point M , and let it come to rest. Through M draw a vertical line MT . The point of

CENTRE OF GRAVITY OF A BODY

A body comprises of several parts and its every part possesses weight.

Weight is the force of attraction between a body and the earth and is proportional to mass of the body.

The weights of all parts of a body can be considered as parallel forces directed towards the Centre of the earth. Therefore, they may be considered into a resultant force whose magnitude is equal to their algebraic sum. If a supporting force, equal and opposite to the resultant, is applied to the body along the line of action of the resultant, the body will be in equilibrium. This line of action will pass through the Centre of Gravity of the body.

Centre of gravity of the body may be defined as the point through which the whole weight of a body may be assumed to act. The Centre of gravity of a body or an object is usually denoted by C.G. or simply by G. The position of C.G depends upon shape of the body and this may or may not necessarily be within the ~~body~~ boundary of the body.

Also the maximum bending moment due to uniformly distributed load is =

$$\frac{wL^2}{8} = \frac{4 \times 1^2}{8} = 0.5 \text{ kNm}$$

$$\therefore 0.5 \text{ k}^2 = 28.25$$

$$L = 7.516 \text{ m}$$

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Example 3: A floor has to carry a load of 12 kN/m² square metre. The floor is supported on rectangular joists each 300 mm x 150 mm and 5 m long. Calculate the distance apart (from the centre to centre) at which joists should be placed so that the maximum stress in the joists should not exceed 8 N/mm².

Solution:

Moment of inertia of each joist

$$I = \frac{bd^3}{12} = \frac{0.1(0.3)^3}{12} = 2.25 \times 10^{-4} \text{ m}^4$$

Maximum B.M (m) to which each joint

$$\frac{W \times x^3}{6EI}$$

$$M = \frac{5I}{5} = \frac{8 \times 2.25 \times 10^4}{\frac{0.30}{2}} = 0.012 \text{ MNm}$$

= 12 kNm — ①

Let x be the spacing, in metres, of the joints from centre to centre. Each joint will share half of the load of the floor between two floor joints on either side. The length of each joint is 5m.

Therefore the total on each joint (if the spacing between two joints is x from centre to centre) will be

$$= 12 \times x \times 5 = 60x \text{ kN}$$

$$\text{Max B.M} = \frac{Wl}{8} = \frac{60x \times 5}{8} \text{ kNm} \quad \text{--- ②}$$

From ① & ② above, we get

$$\frac{12 = 60x \times 5}{8}$$

$$x = \frac{12 \times 8}{60 \times 5} = 0.32 \text{ m}$$

$$x = 320 \text{ mm}$$

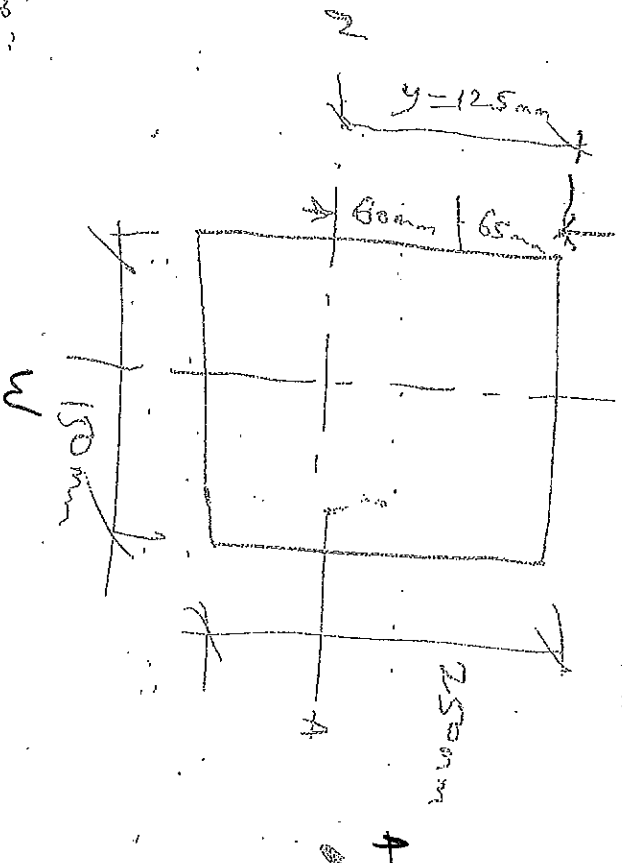
Example ①: A 250 mm (depth) $\times$ 150 mm (width) rectangular beam is subjected to maximum bending moment of 750 kNm as shown in figure ① below.

~~Date: / /~~

(i) The maximum stress in the beam

(ii) If the value of E for the beam material is 200 GPa , find out the radius of curvature for that portion of the beam where the bending is maximum.

(iii) The value of the longitudinal stress at a distance of 65 mm from the top surface of the beam.



Solution:-

Width of the beam, $b = 150 \text{ mm} = 0.15 \text{ m}$

Depth of the beam, $d = 250 \text{ mm} = 0.25 \text{ m}$

Maximum bending moment, $M = 750 \text{ kNm}$

Young's modulus of elasticity, $E = 200 \text{ kN/m}^2$

i) Maximum Stress in the beam

$$\text{Moment of inertia, } I = \frac{bd^3}{12} = \frac{0.15 \times 0.25^3}{12} = 0.0001953 \text{ m}^4$$

Distance of neutral axis (N.A) from the top surface of the beam

$$x = \frac{d}{2} \quad y = \frac{d}{2} = \frac{0.25}{2} = 0.125 \text{ m}$$

Using the relation, $\frac{M}{I} = \frac{\sigma}{y}$ we get

$$\begin{aligned} \sigma &= \frac{M \times y}{I} = \frac{750 \times 10^3 \times 0.125}{0.0001953} \\ &= 4.8 \times 10^8 \text{ N/m}^2 \quad \text{or} \\ &= 480 \text{ MN/m}^2 \end{aligned}$$

Hence, the maximum stress in the beam = 480 MN/m²

$$I = \frac{bd^3}{12} = 0.15 \times 0.25^3$$

(ii) Radius of Curvature, R
using the relation,

$$\frac{M}{I} = \frac{E}{R}, \text{ we get}$$

$$R = \frac{EI}{M} = \frac{200 \times 10^9 \times 0.0001953}{750 \times 10^3} \\ = 52.08 \text{ m}$$

(iii) longitudinal Stress at a distance of 65 mm from top surface of the beam, σ_1

using the relation,

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{\sigma_1}{y}, \text{ we get}$$

$$\sigma_1 = \frac{M \times y}{I}$$

$$= \frac{750 \times 10^3 \times (60 \times 10^{-3})}{0.0001953} \times 10^{-6} \quad (\because y = 125 - 60 = 65 \text{ mm}) \\ = 230.4 \text{ MN/m}^2$$

Example 2: A Symmetrical Section 200 mm deep has a moment of inertia of $2.26 \times 10^{-5} \text{ m}^4$ about its neutral axis. Determine the longest span over which, when simply supported, the beam would carry a uniformly distributed load of 4 kN/m run without the stress due to bending exceeding 125 MN/m^2 .

Solution:

Depth of the Symmetrical Section, $d = 200 \text{ mm}$

Moment of inertia about neutral axis, $I = 2.26 \times 10^{-5} \text{ m}^4$

Uniformly distributed load, $W = 4 \text{ kN/m run}$

Maximum bending stress, $\sigma = 125 \text{ MN/m}^2$

Longest span, $L =$

using the relation

$$\frac{M}{I} = \frac{\sigma}{y}, \text{ we have}$$

$$M = \frac{\sigma I}{y} = \frac{125 \times 10^6 \times 2.26 \times 10^{-5}}{0.2/2}$$

$$= 28.25 \times 10^3 \text{ Nm}$$

$$= 28.25 \text{ kNm}$$